

Foundations of AI and Robotics

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Definition and scope of AI and roboticsHistory and evolution of intelligent robots

PDF Documents

PDF Document 1: 1747042832874-AI_Robotics_Course_Units.pdf

This PDF document is included in the unit materials.

Unit 1: Foundations of AI and Robotics

- Definition and scope of AI and robotics
- History and evolution of intelligent robots
- Differences and synergies between AI and robotics
- Real-world applications (autonomous cars, drones, industry)
- Challenges in merging AI with robotics

Unit 2: Robotic Hardware Architecture

- Types of robots (mobile, industrial, humanoid)
- Sensors (IR, ultrasonic, LIDAR, vision sensors)
- Actuators and motors (servo, stepper, DC)
- Microcontrollers and single-board computers (Arduino, Raspberry Pi)
- Power systems and control boards

Unit 3: Perception and Computer Vision

- Basics of image processing (filtering, edge detection)
- Camera calibration and object recognition
- Depth estimation and stereo vision
- OpenCV implementation in Python
- Introduction to SLAM (Simultaneous Localization and Mapping)

Unit 4: Machine Learning & Deep Learning for Robotics

- Supervised, unsupervised, and reinforcement learning
- Feature extraction and data preprocessing
- Neural networks (ANNs, CNNs, RNNs)
- Transfer learning for robotics
- Real-time AI on edge devices

Unit 5: Navigation, NLP, ROS, and Ethics

- Pathfinding algorithms: A*, Dijkstra, RRT
- Sensor fusion and autonomous navigation
- NLP for voice-controlled robots
- Basics of ROS (Robot Operating System)
- Ethics, safety, and future trends in AI robotics